

Worksheet 1: The Root System

NCERT Class 11 Biology — Chapter 5: Morphology of Flowering Plants

Student Name: _____	Roll No.: _____	Section: _____
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Section A: Multiple Choice Questions (Select the single best answer)

- Which of the following root systems develops directly from the elongation of the embryonic radicle? [____]
A) Fibrous root system
B) Tap root system
C) Adventitious root system
D) Foliar root system
- Root hairs, which carry out maximal absorption of water and minerals, arise exclusively from: [____]
A) Root cap
B) Region of meristematic activity
C) Region of elongation
D) Region of maturation
- In monocots like Wheat, the primary root is short-lived and replaced by roots originating from: [____]
A) The upper nodes of the stem
B) The base of the stem
C) The leaf petiole
D) The radicle itself
- Which of the following is an example of an adventitious root system? [____]
A) Mustard
B) Wheat
C) Banyan tree / Monstera
D) Gram

Section B: Fill in the Blanks & True/False

- In aquatic plants, the protective root cap is absent and replaced by _____.
- Root hairs are _____ (unicellular / multicellular) tubular outgrowths of epidermal cells.
- State True or False: Cells in the region of meristematic activity have thick cell walls and sparse protoplasm. [____]
- State True or False: The region of elongation is primarily responsible for the growth of the root in length. [____]

Section C: Complete the Zone Characteristics Table

Zone / Region	Position / Location	Key Cellular Characteristics & Primary Function
Root Cap	Terminal Apex	_____ _____
Meristematic Activity	Few mm above root cap	_____ _____
Region of Elongation	Proximal to meristem	_____ _____

Region of Maturation	Proximal to elongation	_____

Section D: Short Answer & Analytical Questions

9. Differentiate between Tap Root System and Fibrous Root System on the basis of origin, lifespan of primary root, and plant type.

10. State four primary physiological and mechanical functions performed by the root system in angiosperms.

Lecturer Signature: _____